Machine Intelligence 1

1.2 Connectionist Neurons

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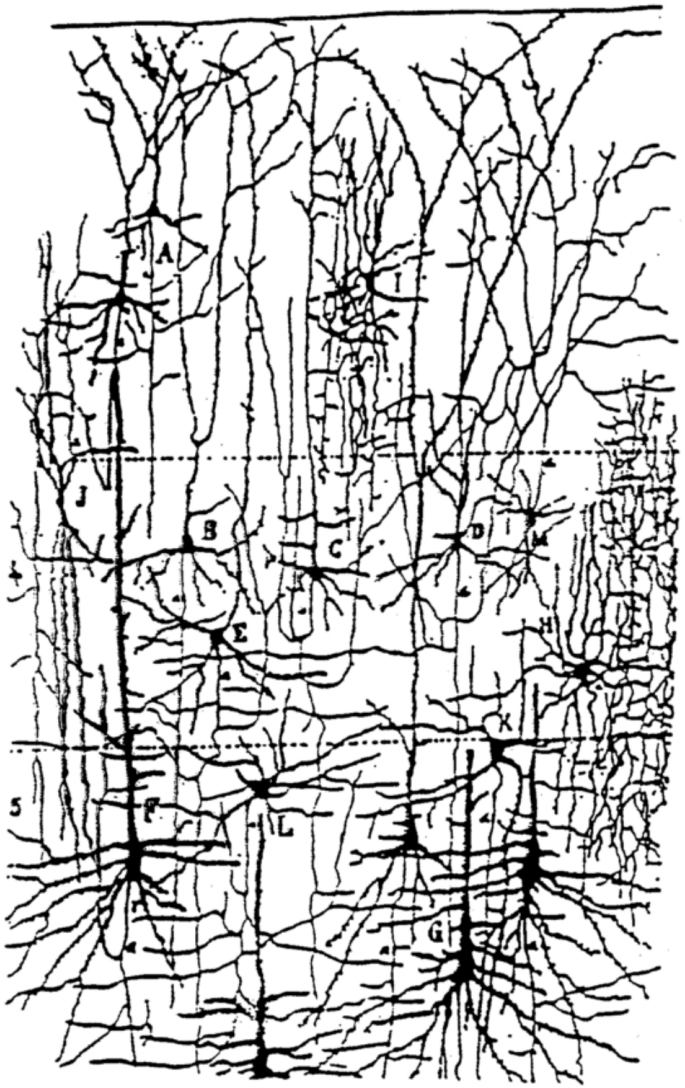
Fachgebiet Neuronale Informationsverarbeitung (NI)

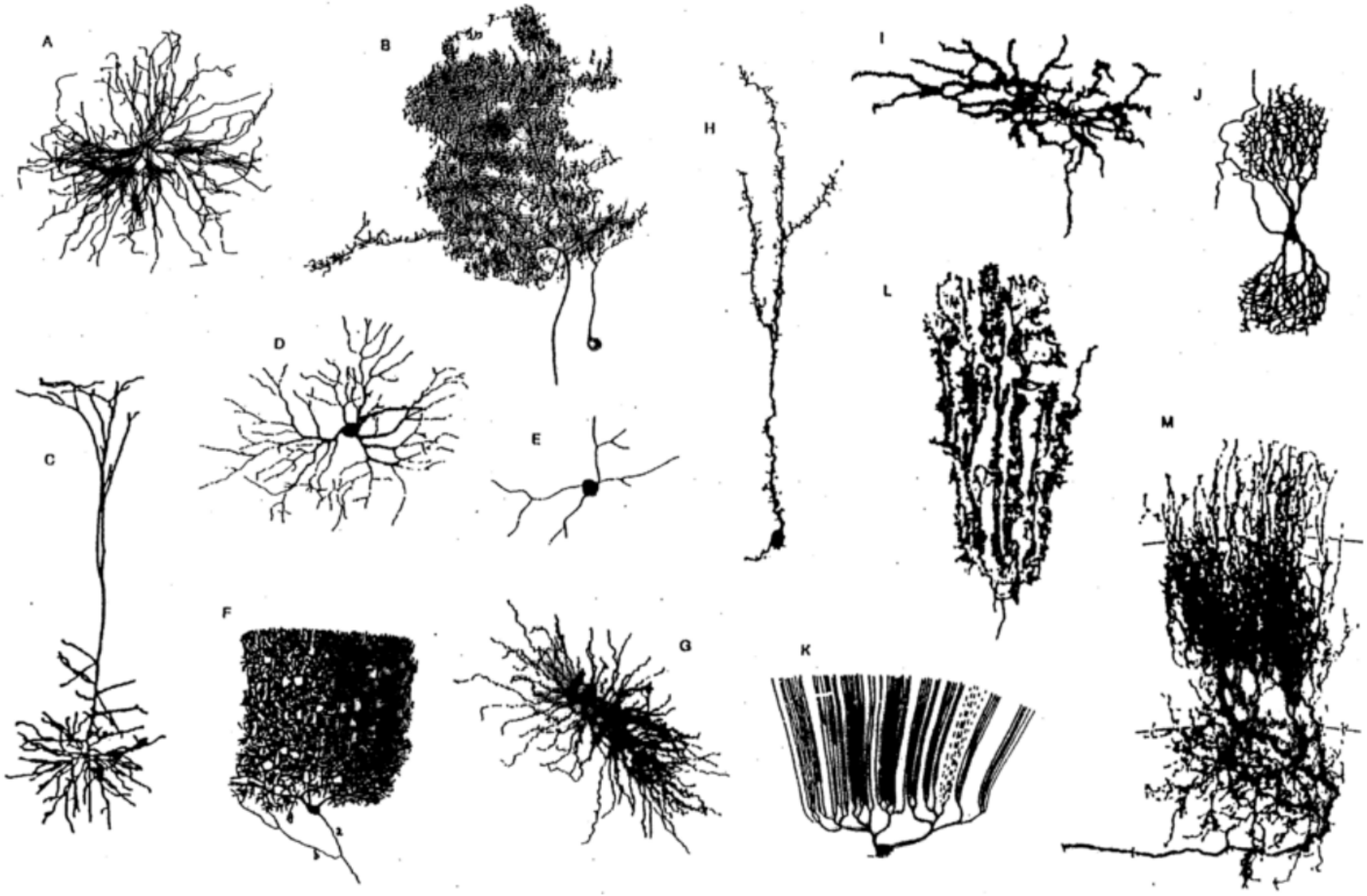
WS 25/26

Neural Networks



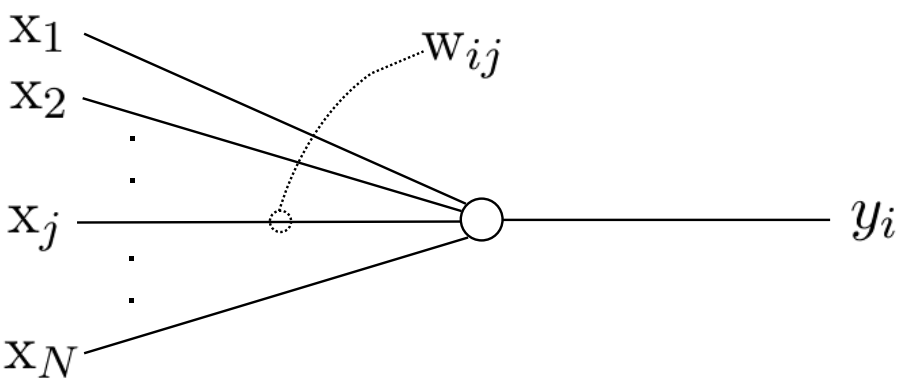
1.2.1 Input-Output Relationship

Biological neurons



Connectionist neurons

Input-output relationship: linear filter with a static non-linearity



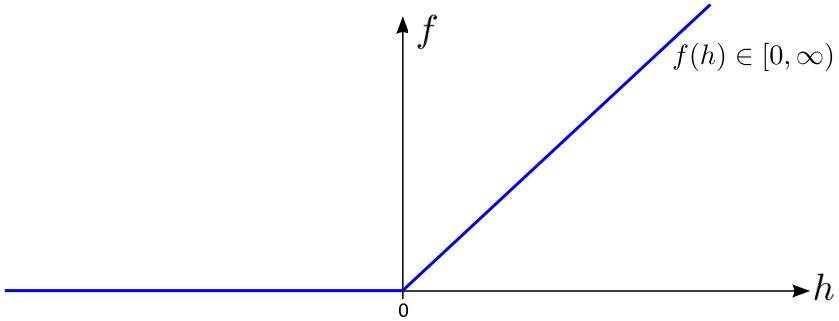
$$\rightarrow \quad y_i = f\Big(\underbrace{\sum_j \mathrm{w}_{ij}\mathrm{x}_j - \theta_i}_{h_i}\Big)$$

$\underline{\mathbf{x}}$:	input vector with components x_j
y_i :	scalar output of neuron i
$\underline{\mathbf{w}}_i$:	weight vector of neuron i with components w_{ij}
θ_i :	threshold of neuron i
h_i :	total input of neuron i
f :	transfer function

Examples: rate neurons, mean-field approximation, receptive field models, etc.

Typical transfer functions (1)

rectified linear function
$f(h) = \max(0, h)$



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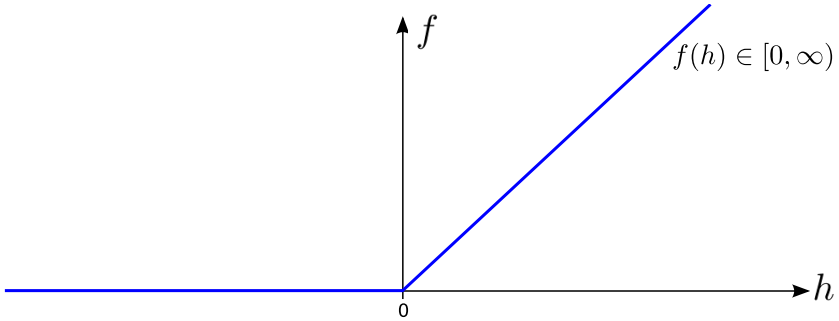
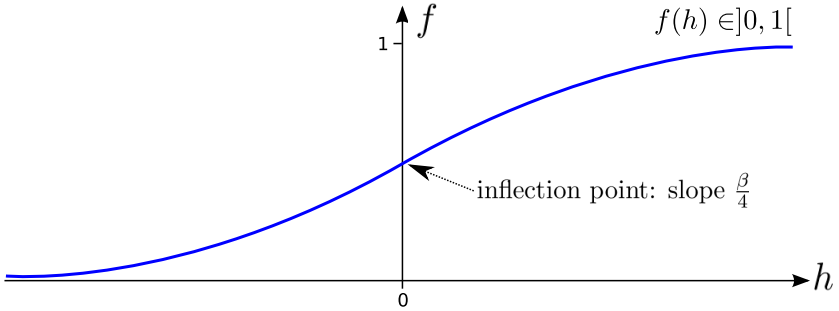


logistic function
$f(h) = \frac{1}{1+\exp(-\beta h)}$



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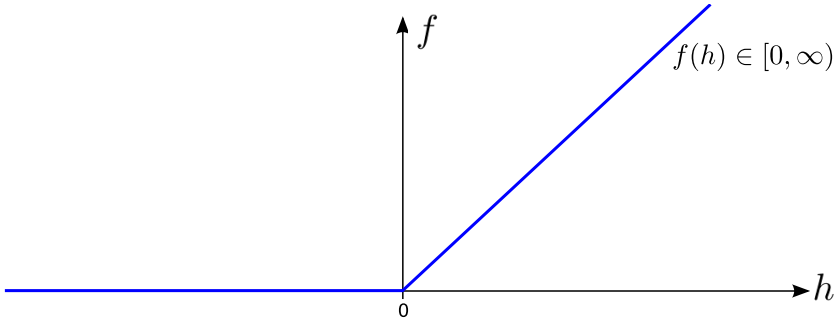

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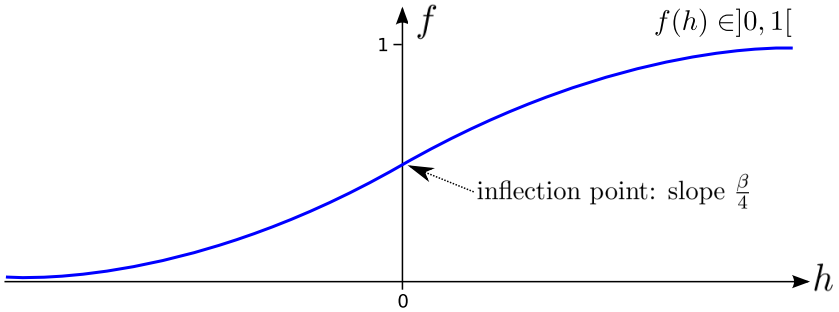
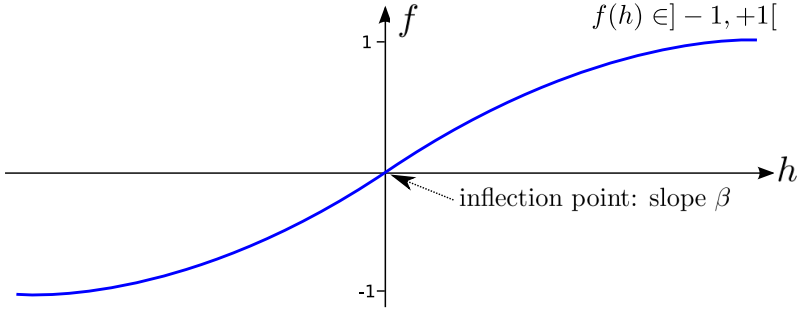


hyperbolic tangent
$f(h) = \tanh(\beta h)$



Typical transfer functions (2)

$$
\begin{aligned}
\frac{1}{1+e^{-h}} &= \frac{e^{\frac{h}{2}}}{e^{\frac{h}{2}}+e^{-\frac{h}{2}}} \\
&= \frac{1}{2}\left\{ \underbrace{\frac{e^{\frac{h}{2}}-e^{-\frac{h}{2}}}{e^{\frac{h}{2}}+e^{-\frac{h}{2}}}}_{=\tanh\frac{h}{2}} + \underbrace{\frac{e^{\frac{h}{2}}+e^{-\frac{h}{2}}}{e^{\frac{h}{2}}+e^{-\frac{h}{2}}}}_{=1} \right\} \\
&= \frac{1}{2}\left(\tanh\frac{h}{2}+1\right)
\end{aligned}
$$

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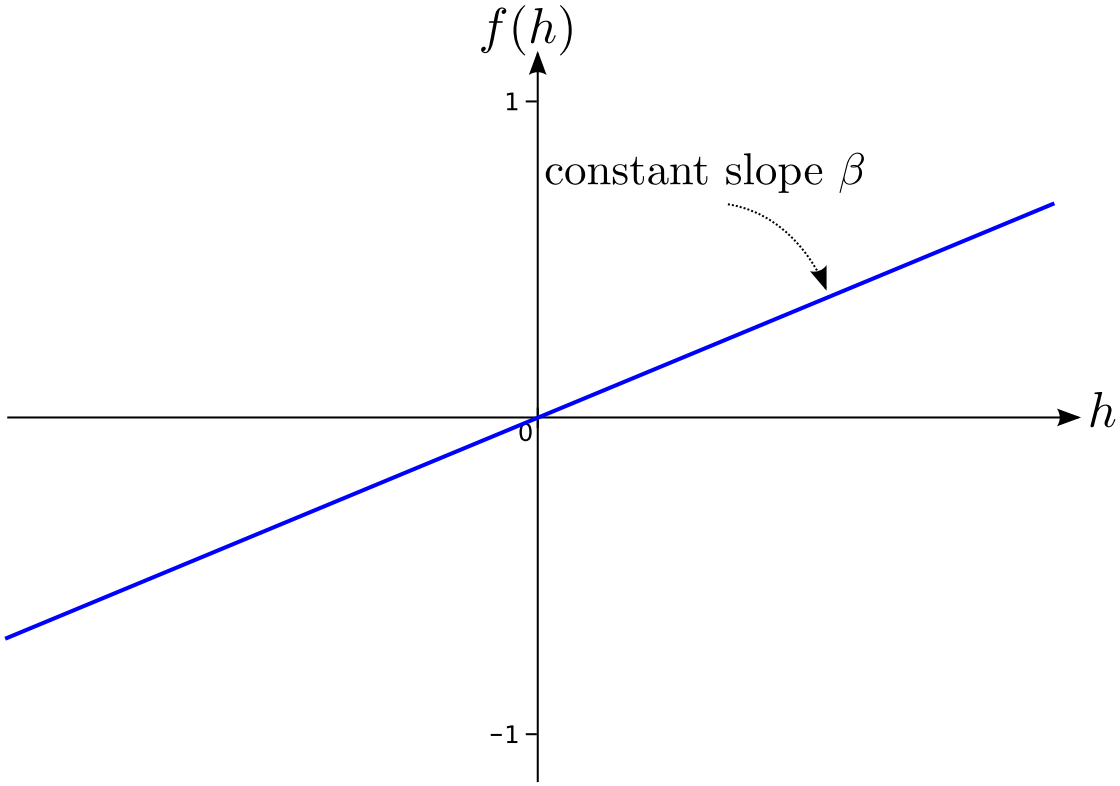
&= \frac{1}{2}\left(\tanh\frac{h}{2}+1\right)
\end{aligned}
$$

transformation

1. scale all input weights w_{ij} or slope parameter β by 2
2. shift output by -1
3. scale output by 2

Special transfer functions

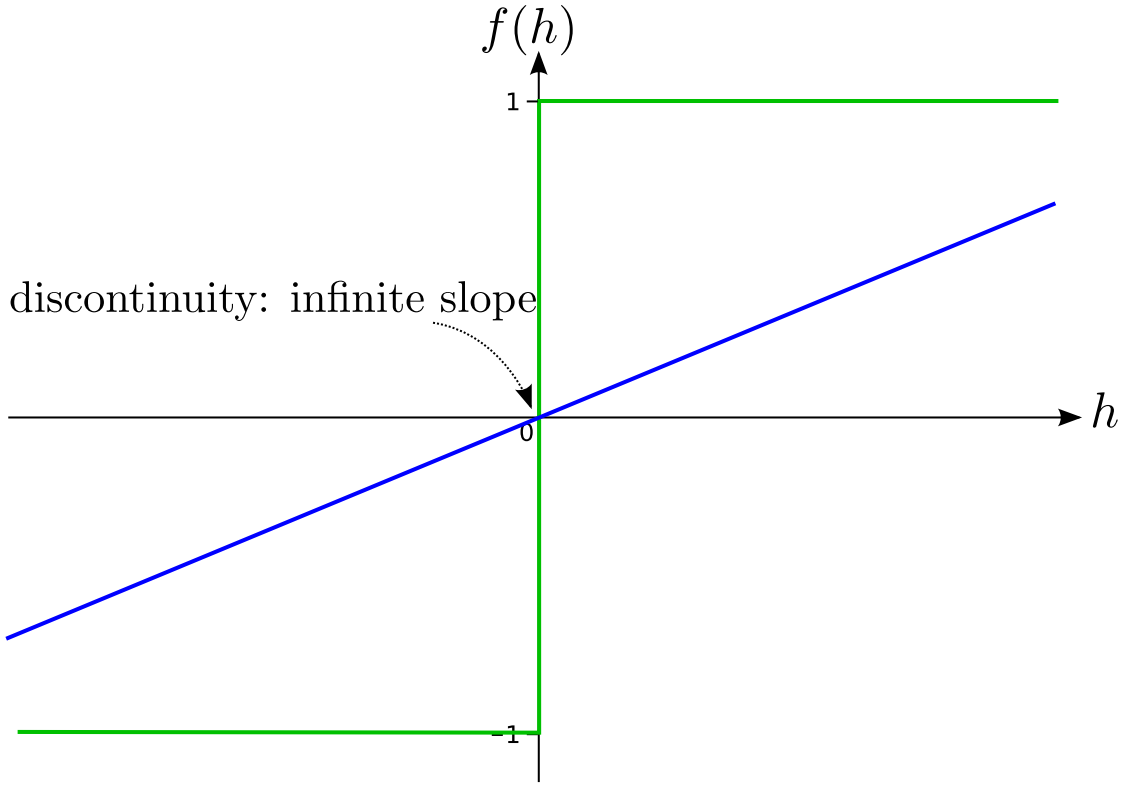
Linear neuron: $f(h) = \beta h$ extraction of linear features



Special transfer functions

Linear neuron: $f(h) = \beta h$ extraction of linear features

Binary neuron: $f(h) = \mathrm{sign}(h)$ classification

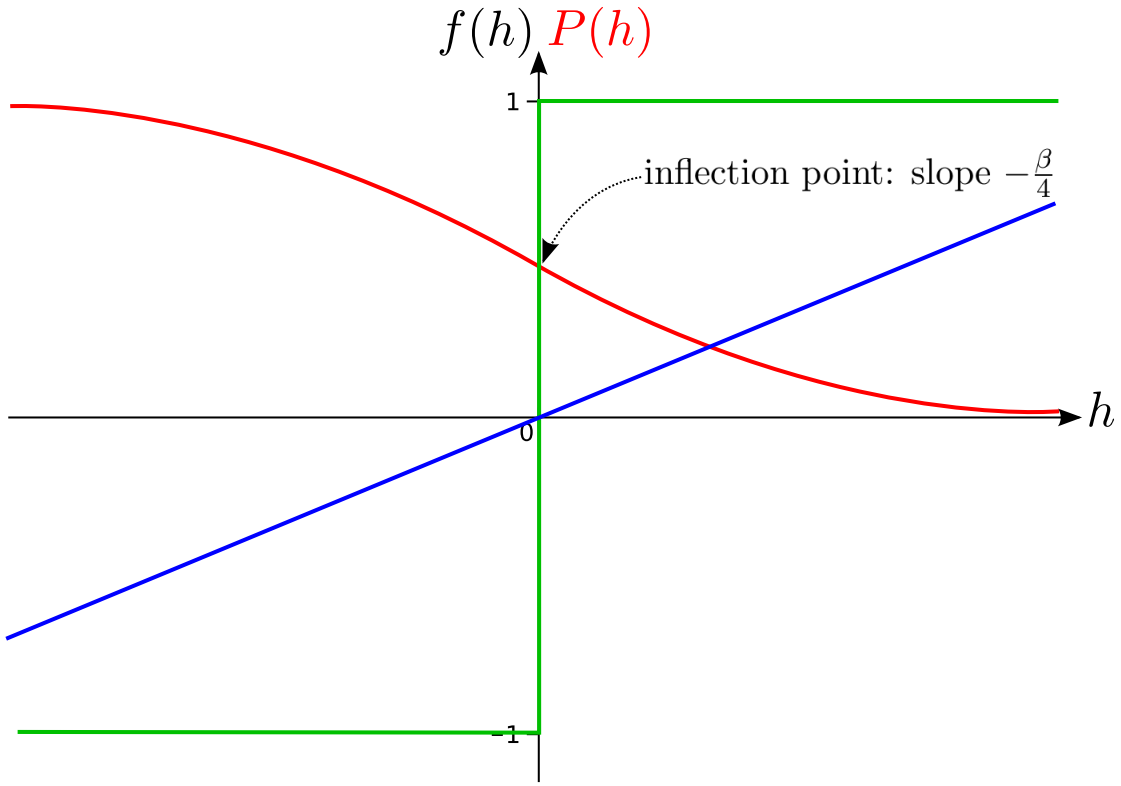


Special transfer functions

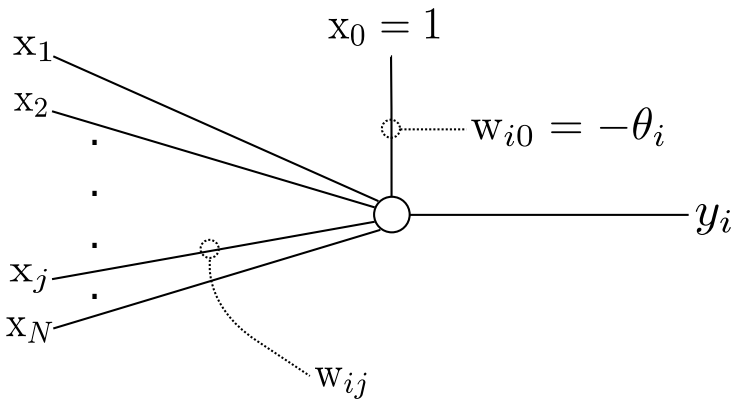
Linear neuron: $f(h) = \beta h$ extraction of linear features

Binary neuron: $f(h) = \mathrm{sign}(h)$ classification

Stochastic binary neuron: $P(y \to -y) = \frac{1}{1+\exp(\beta y h)}$ noise parameter β



Shortcut notation for threshold / bias



$$y_i = f\left(\sum_{j=0}^{N} w_{ij} x_j\right) = f\left(\underline{\mathbf{w}}_i^\top \underline{\mathbf{x}}\right)$$

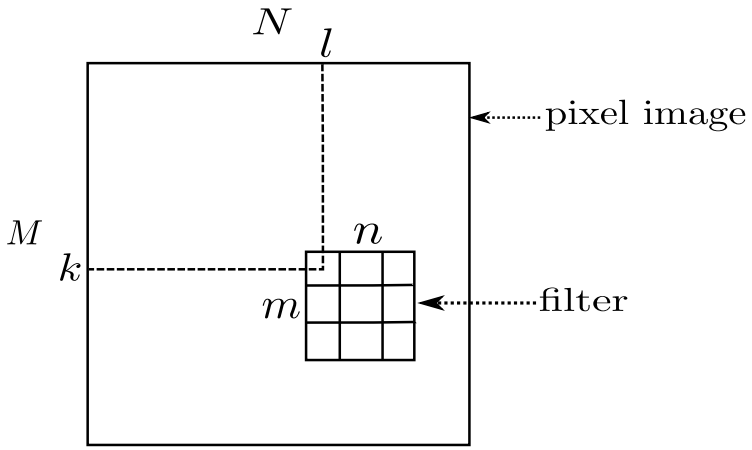
Note on notation

$\underline{\mathbf{w}}$ will be used for $\begin{pmatrix} w_1 \\ \vdots \\ w_N \end{pmatrix}$ as well as for $\begin{pmatrix} w_0 \\ w_1 \\ \vdots \\ w_N \end{pmatrix}$

$\underline{\mathbf{x}}$ will be used for $\begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix}$ as well as for $\begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_N \end{pmatrix}$

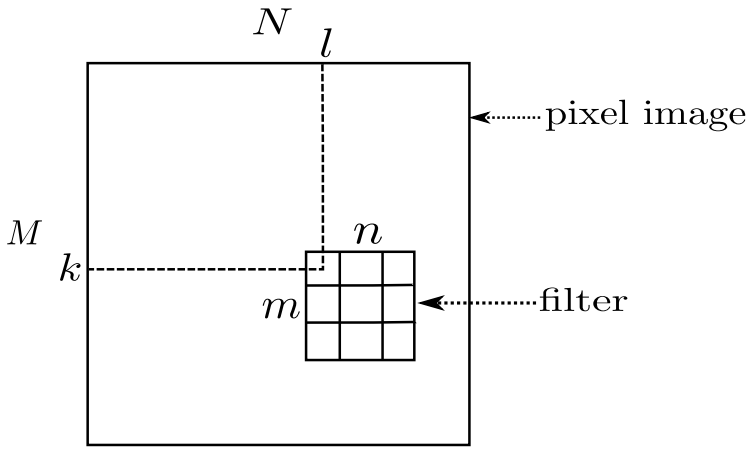
1.2.2 Feature Extraction and Evaluation

Linear filters and feature extraction



$$\underbrace{y_{kl}}_{\text{strength of feature}} = \sum_{i,j=0}^{m,n} \underbrace{\mathrm{w}_{ij}}_{\text{filter coefficients}} \; \underbrace{\mathrm{x}_{(k+i)(l+j)}}_{\text{pixel value}}$$

Linear filters and feature extraction



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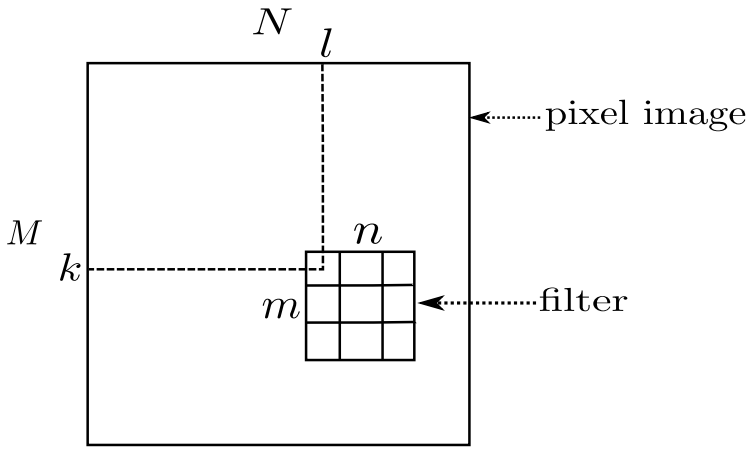
Example filters $\underline{\mathbf{w}}$ for points and edges

-1	-1	-1
-1	+8	-1
-1	-1	-1

point filter

Linear filters and feature extraction



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Example filters $\underline{\mathbf{w}}$ for points and edges

-1	-1	-1
-1	+8	-1
-1	-1	-1

point filter

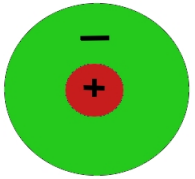
-1	0	+1
−2	0	+2
-1	0	+1

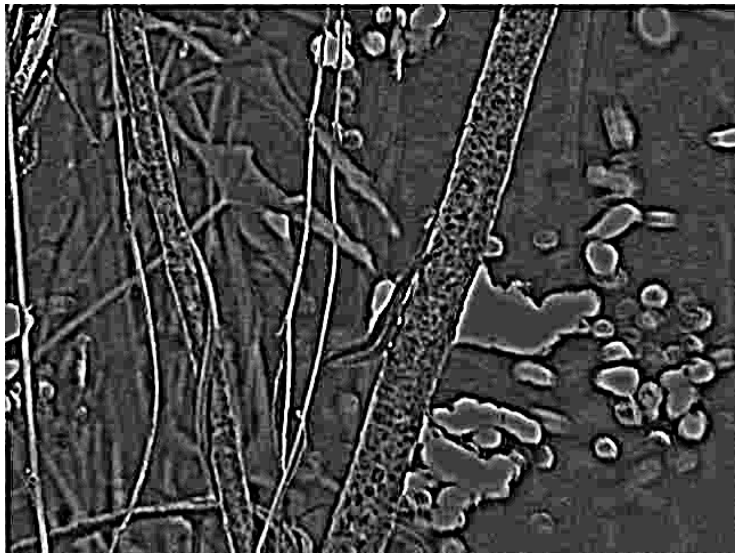
+1	+2	+1
0	0	0
-1	−2	-1

edge filter (Sobel filter)

⇒ $\underline{\mathbf{w}}$ describes a feature and is often called receptive field.

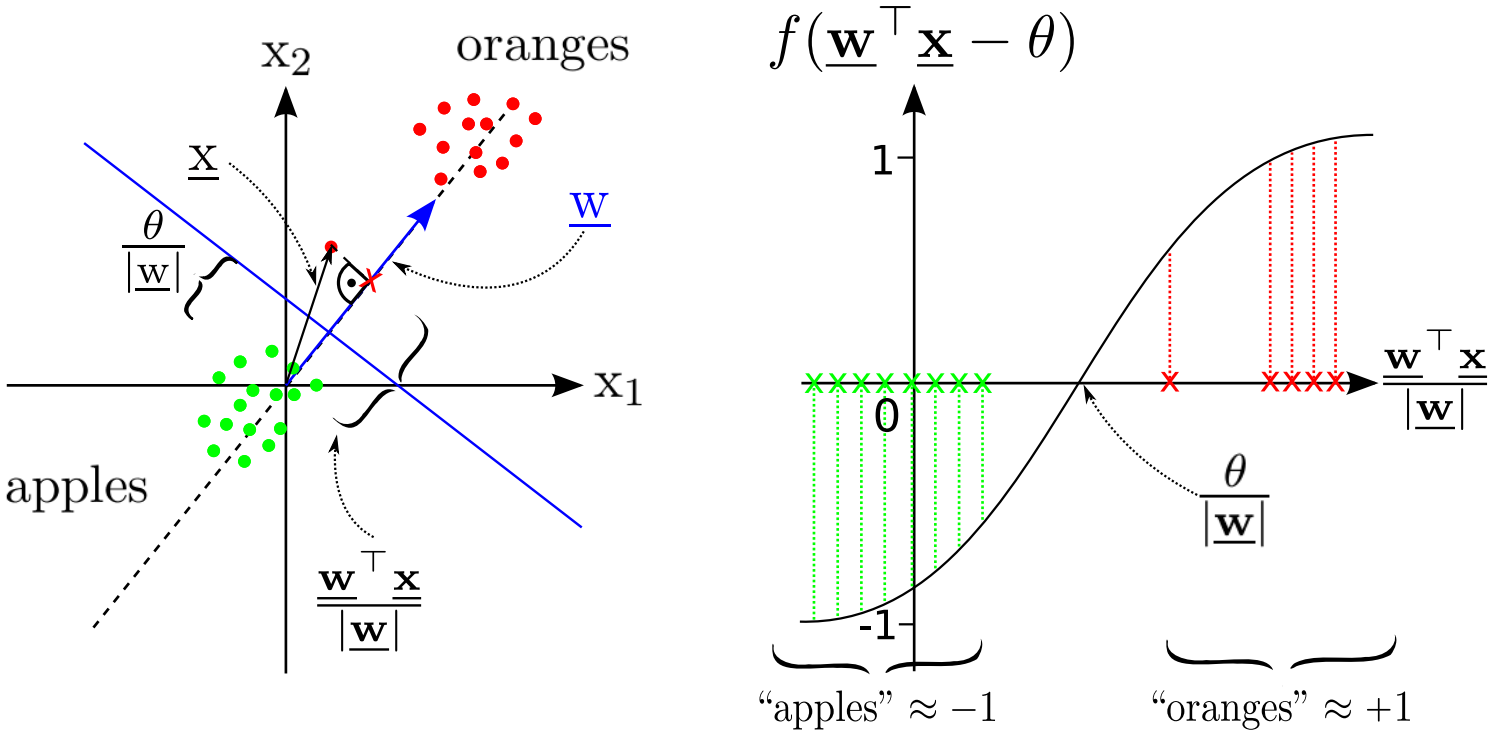
Center-surround filters





from: http://www.union.edu/academic_depts/bioengineering/visual_motion/visual_tour/center_surround.php

Extraction / evaluation of features



$$\underbrace{\underline{x} \longrightarrow \underline{w}^\top \underline{x}}_{\text{feature extraction}} \underbrace{\longrightarrow f(\underline{w}^\top \underline{x} - \theta)}_{\text{evaluation}}$$

$\Rightarrow$ Partitioning of feature space into two half-spaces